Fusion of Plant Disease Classification: Betel Leaf Disease Diagnosis Dataset

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***Abstract*—Betel leaf is an economically and culturally significant crop widely cultivated in South and Southeast Asia. However, it is highly susceptible to diseases such as leaf rot and leaf spot, which adversely affect crop yield and quality. Early and accurate detection of these diseases is essential for effective crop management and minimizing economic losses.**

In this study, a hybrid deep learning framework is proposed for automatic betel leaf disease classification. The model integrates EfficientNet-B0, a convolutional neural network, with Swin-Tiny, a transformer-based architecture, using feature-level fusion. This combination enables the extraction of both local texture features and global spatial dependencies from leaf images. To address dataset imbalance, oversampling and weighted random sampling techniques are employed.

The proposed model achieves a classification accuracy of 97.07%, a macro F1-score of 0.9597, and a macro AUC of 0.9985. Experimental results demonstrate that the fusion-based approach significantly outperforms conventional single-model methods. Furthermore, Grad-CAM visualization is utilized to enhance model interpretability by highlighting disease-relevant regions. The proposed system provides a reliable and efficient solution for real-world betel leaf disease diagnosis.

***Index Terms*—Betel leaf, disease detection, deep learning, CNN, transformer, feature fusion**

I. Introduction

Betel leaf (Piper betle L.) is an important agricultural crop widely used for medicinal, cultural, and commercial purposes, particularly in countries such as Bangladesh, India, and Sri Lanka. It plays a vital role in local economies and traditional practices. However, betel leaf cultivation is significantly affected by various diseases, including leaf rot and leaf spot, which degrade leaf quality and reduce overall production. These diseases often spread rapidly under favorable environmental conditions, leading to substantial financial losses for farmers.

Traditionally, disease detection in betel leaves relies on manual inspection by farmers or agricultural experts. This approach is time-consuming, subjective, and prone to human error, especially in the early stages of disease development when symptoms are less visible. Therefore, there is a growing need for automated, accurate, and efficient disease detection systems.

Recent advancements in artificial intelligence, particularly in deep learning, have revolutionized image-based plant disease detection. Convolutional Neural Networks (CNNs) have demonstrated strong performance in extracting local features such as texture, color, and disease patterns from leaf images. However, CNN-based models are limited in capturing long-range dependencies and global contextual information. On the other hand, transformer-based architectures, such as Vision Transformers (ViT) and Swin Transformers, are capable of modeling global relationships within images but may lack fine-grained local feature extraction.

To overcome these limitations, hybrid approaches that combine CNNs and transformers have gained attention. Fusion-based methods allow the integration of complementary features from multiple architectures, leading to improved performance. However, such approaches are still underexplored in the context of betel leaf disease classification.

In this study, we propose a hybrid deep learning framework that combines EfficientNet-B0 and Swin-Tiny using feature-level fusion. The EfficientNet-B0 model captures detailed local features, while the Swin-Tiny transformer extracts hierarchical global features. These features are fused into a unified representation for classification. Additionally, techniques such as dataset balancing and advanced data augmentation are employed to enhance model robustness.

The main contributions of this work are as follows:

- A hybrid CNN–Transformer architecture for betel leaf disease classification.
- A feature-level fusion strategy combining local and global representations.
- An effective dataset balancing approach using oversampling and weighted sampling.
- Comprehensive evaluation using multiple metrics and Grad-CAM-based explainability.

The proposed approach aims to provide a reliable and efficient tool for early disease detection, ultimately supporting farmers in improving crop management and productivity.

II. Related Work

Plant disease detection has been extensively studied using image processing, machine learning, and deep learning techniques. Early approaches primarily relied on handcrafted feature extraction and classical classifiers. For example, Dey et al. [21] proposed an image-processing-based system for detecting leaf rot disease in betel vine by segmenting infected regions and estimating the affected area. Similarly, Zeba et al. [22] applied Gaussian mixture model-based segmentation and an SVM classifier for betel leaf disease classification. While these methods demonstrated the feasibility of automated

detection, their reliance on handcrafted features limits their ability to generalize across complex field conditions.

Traditional machine learning approaches have also been explored for betel leaf disease classification. David and Mukunthan [2] utilized GLCM texture features combined with an Extreme Learning Machine (ELM) classifier, achieving strong performance. Additionally, Sharma et al. [14] and Man et al. [13] demonstrated that preprocessing, segmentation, and machine learning classifiers can provide reliable baselines. However, these approaches depend heavily on manual feature engineering and struggle to capture complex visual patterns.

With the advancement of deep learning, Convolutional Neural Networks (CNNs) have become the dominant approach for plant disease detection. Lakshmi and Nickolas [9] introduced one of the early CNN-based models for betel leaf disease detection, demonstrating the effectiveness of deep feature learning. Malek et al. [23] compared multiple deep learning models and showed that transfer learning significantly improves classification accuracy, with InceptionV3 achieving superior performance. Similarly, Devi and Raj [3] applied CNN and VGG16 models with data augmentation and reported improved results compared to traditional approaches. Despite their effectiveness, these methods are limited to single-model architectures.

Lightweight and application-oriented CNN models have also been proposed. A depth-wise separable CNN model demonstrated efficient performance for early betel leaf disease detection, while Nagaraj et al. [24] explored YOLOv3-based detection integrated into a mobile application framework. Although these approaches are suitable for real-time deployment, they often lack comprehensive evaluation on large-scale datasets.

Ensemble learning has shown promising improvements in plant disease classification. Shovon et al. [1] proposed a deep ensemble model combining InceptionResNetV2, EfficientNetV2L, and Xception, achieving high accuracy on plant disease datasets, including betel leaf. Similarly, Hridoy et al. [5] developed an ensemble-based framework for multi-plant disease classification and reported very high performance. While ensemble methods improve robustness, they typically operate at the prediction level rather than integrating feature representations.

Attention mechanisms have further enhanced deep learning models by enabling them to focus on disease-relevant regions. Tabassum and Ahamed [4] introduced a CBAM-based attention model combined with Grad-CAM for explainability, achieving improved classification performance. However, the approach still relies on a single backbone model and does not exploit multi-model feature integration.

Transformer-based architectures have recently gained attention for their ability to capture global contextual information. Kusuma and Jothi [10] compared CNN models with Vision Transformers for betel leaf disease detection and demonstrated competitive performance across both approaches. Gayakwad et al. [11] further explored transfer learning and transformer-based models on large-scale betel datasets. Despite their advantages, most existing studies apply CNNs or transformers independently rather than combining them.

Feature fusion techniques have shown strong potential in improving classification performance. Fan et al. [6] combined deep learning features with handcrafted features and achieved significant performance improvements. Similarly, Yao et al. [15] explored multi-prediction deep learning frameworks for plant disease classification, highlighting the benefits of richer feature representations. However, such fusion strategies are not widely applied to betel leaf datasets.

Dataset development has also contributed to progress in this field. Hridoy et al. [8] introduced a comprehensive betel leaf dataset collected under natural conditions and expanded through augmentation, providing a benchmark for future research. Larger datasets enable better generalization but are often limited to image-only modalities.

Several survey studies provide broader insights into plant disease detection. Dhingra et al. [18] reviewed classical image processing techniques, while Vishnoi et al. [12], Bhargava et al. [16], and Goyal and Punam [20] analyzed machine learning and deep learning approaches. More recently, Yang et al. [7] highlighted the importance of multimodal fusion, and Wu et al. [17] proposed a graph-based multimodal fusion framework for plant disease diagnosis. These studies emphasize the growing importance of combining multiple representations for improved performance.

Despite significant advancements, several limitations remain. Most existing methods rely on single-model architectures or simple ensemble strategies. Feature-level fusion between heterogeneous models, particularly CNNs and transformers, is still underexplored in betel leaf disease classification. Additionally, many studies do not fully utilize explainability techniques or robust imbalance-handling strategies.

To address these limitations, this study proposes a hybrid CNN–Transformer fusion framework that combines EfficientNet-B0 and Swin-Tiny. By integrating local texture features and global contextual information through feature-level fusion, the proposed method aims to improve classification accuracy, robustness, and interpretability for betel leaf disease detection.

III. Methodology

This section presents the proposed hybrid deep learning framework for betel leaf disease classification. The overall pipeline includes dataset preparation, preprocessing, class imbalance handling, feature extraction using complementary architectures, feature-level fusion, classification, and interpretability analysis.

A. Overview of the Proposed Framework

The proposed model combines a Convolutional Neural Network (CNN) and a transformer-based architecture to capture both local and global features. EfficientNet-B0 extracts fine-grained local features such as texture and lesions, while Swin-Tiny captures global contextual relationships using self-attention mechanisms. The extracted features are fused and passed through a classification head for final prediction.

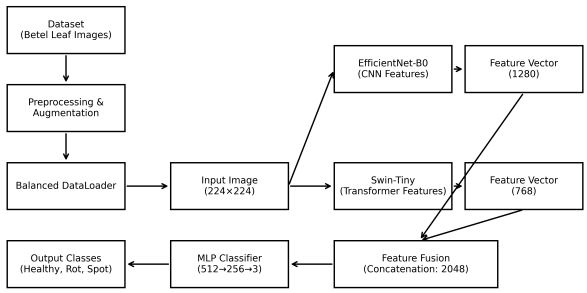

Fig. 1. Overall architecture of the proposed hybrid CNN–Transformer fusion model.

Fig. 2. Sample images from each class: Healthy_Leaf, Leaf_Rot, and Leaf_Spot.

B. Dataset Description

The dataset used in this study consists of both original and augmented betel leaf images categorized into three classes: *Healthy_Leaf*, *Leaf_Rot*, and *Leaf_Spot*. The original dataset contains 2037 images, while the augmented dataset expands the total to 10,185 images using transformations such as flipping, rotation, brightness adjustment, and contrast enhancement.

Fig. 2 shows representative samples from each class. Fig. 3 and Fig. 4 illustrate the dataset distribution across classes.

C. Data Preprocessing and Augmentation

All images are resized to 224×224 pixels to match the input requirements of EfficientNet-B0 and Swin-Tiny. Pixel values are normalized using ImageNet mean and standard deviation.

In addition to the pre-generated augmented dataset, real-time data augmentation techniques are applied during training, including random horizontal and vertical flipping, rotation, color jittering, grayscale transformation, and random erasing. These techniques improve generalization and robustness under varying environmental conditions.

D. Handling Class Imbalance

The dataset exhibits class imbalance, which can negatively impact model performance. To address this, oversampling of minority classes is performed, and a WeightedRandomSampler is used during training to ensure balanced sampling across all classes. This approach improves model stability and macro-level performance.

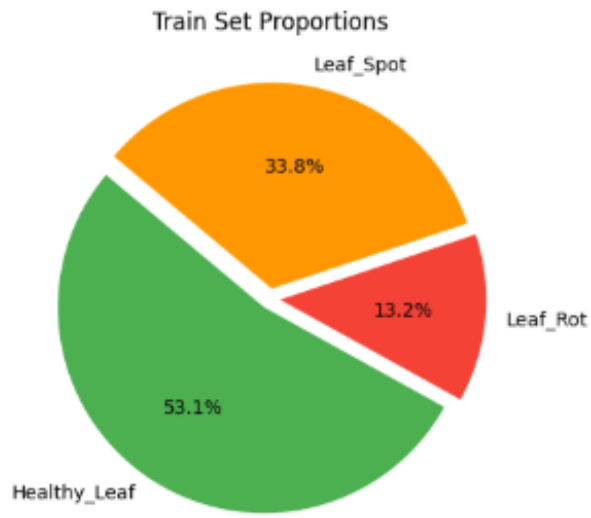

Fig. 3. Pie chart showing class-wise distribution of the dataset.

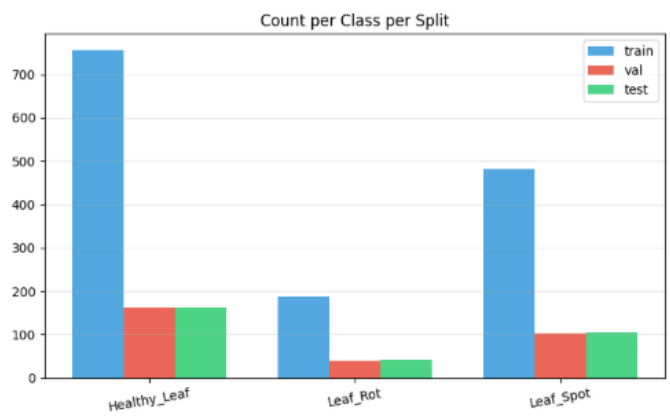

Fig. 4. Bar chart representing the number of images in each class.

E. Feature Extraction

1) EfficientNet-B0 Branch: EfficientNet-B0 is used as the CNN backbone due to its efficiency and strong performance. It extracts local features such as edges, textures, and disease-specific patterns. The output feature vector has a dimensionality of 1280.

2) Swin-Tiny Branch: Swin-Tiny is a transformer-based architecture that captures long-range dependencies and global contextual information using shifted window attention. The output feature vector from this branch has a dimensionality of 768.

F. Feature Fusion

The feature vectors from both branches are concatenated to form a unified representation:

$$F_{fusion} = [F_{cnn}, F_{swin}] \tag{1}$$

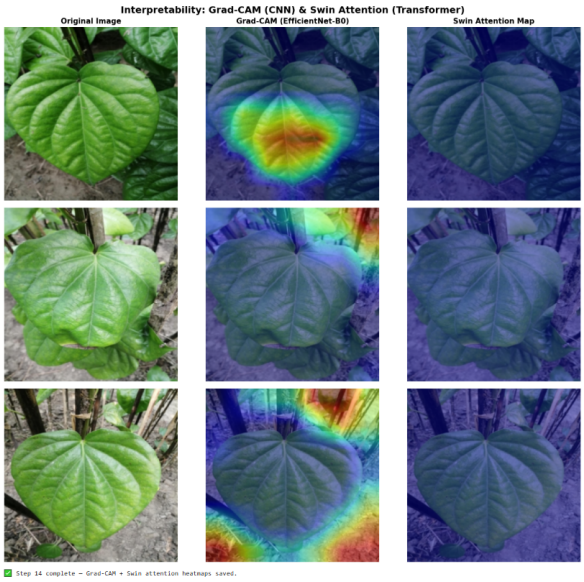

Fig. 5. Grad-CAM visualization highlighting important regions influencing predictions.

The fused feature vector has a dimensionality of 2048, combining local and global information for improved classification performance.

G. Classification Head

The fused features are passed through a multi-layer perceptron (MLP) consisting of:

- Fully connected layer (2048 $\rightarrow$ 512)
- Fully connected layer (512 $\rightarrow$ 256)
- Output layer (256 $\rightarrow$ 3)

Each layer uses GELU activation, batch normalization, and dropout to improve convergence and reduce overfitting.

H. Training Strategy

The model is trained using a two-stage strategy:

- **Stage 1:** Backbone networks are frozen, and only the classification head is trained.
- **Stage 2:** The entire network is fine-tuned for improved performance.

The training configuration includes:

- Loss function: CrossEntropyLoss with label smoothing
- Optimizer: AdamW
- Learning rate scheduling for stable convergence
- Evaluation metric: Macro F1-score

The best model is selected based on validation Macro F1-score to ensure balanced performance across all classes.

I. Model Interpretability

To improve transparency, Grad-CAM is applied to visualize important regions contributing to the model's predictions.

The visualization confirms that the model focuses on disease-relevant regions such as lesions and discoloration, enhancing interpretability and trustworthiness.

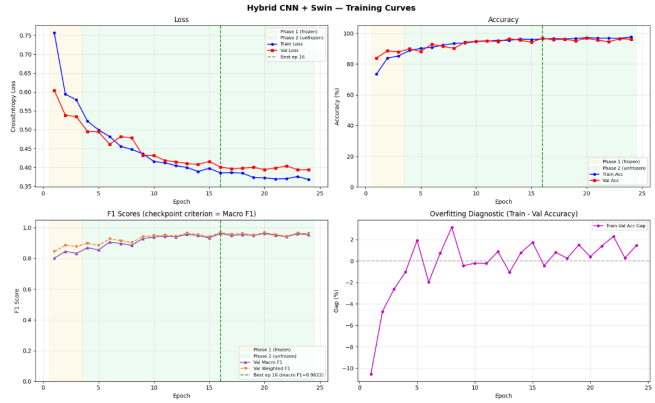

Fig. 6. Training performance showing loss, accuracy, and F1-score across epochs.

IV. Results and Discussion

This section presents a comprehensive evaluation of the proposed hybrid CNN–Transformer model for betel leaf disease classification. The performance is evaluated using multiple metrics including accuracy, Macro F1-score, Weighted F1-score, AUC-ROC, and Precision-Recall (PR) analysis.

A. Overall Performance

The proposed model achieves strong performance across all evaluation metrics:

- Accuracy: 97.07%
- Macro F1-score: 0.9597
- Weighted F1-score: 0.9708
- Macro AUC-ROC: 0.9985
- Macro Average Precision: 0.9950

These results indicate that the model performs consistently across all classes, including minority classes, demonstrating the effectiveness of the fusion-based architecture.

B. Training Performance and Convergence Analysis

The training behavior is illustrated in Fig. 6. The curves show stable convergence with minimal gap between training and validation performance, indicating good generalization and absence of overfitting.

C. ROC Curve Analysis

The ROC curve is presented in Fig. 7. The model achieves a macro AUC of 0.9985, indicating excellent discriminative capability.

D. Precision–Recall Analysis

The PR curve is shown in Fig. 8. The high area under the curve confirms strong performance under class imbalance conditions.

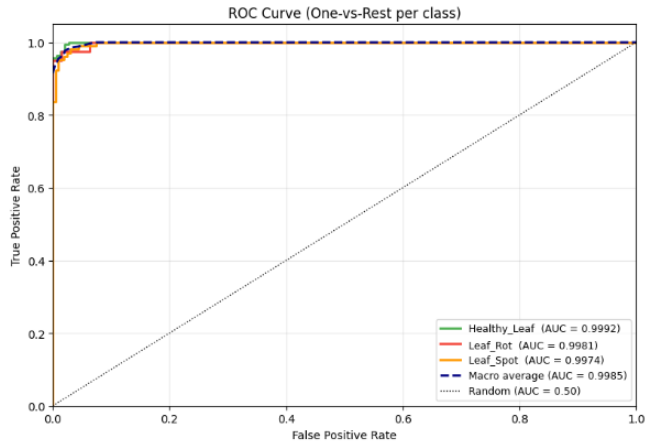

Fig. 7. ROC curve demonstrating high class separability.

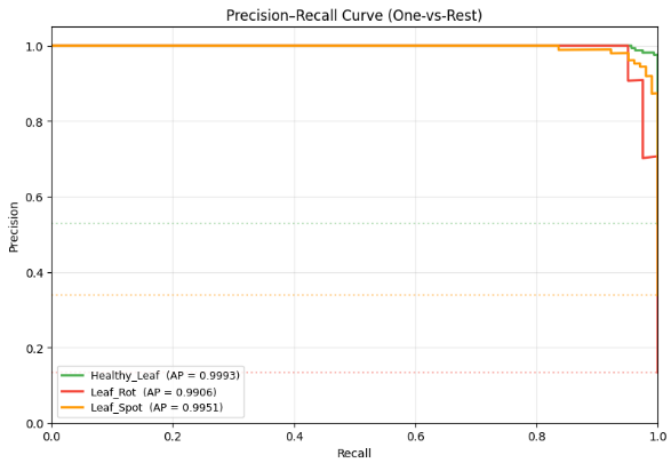

Fig. 8. Precision–Recall curve showing high precision and recall across classes.

E. Confusion Matrix Analysis

The normalized confusion matrix is shown in Fig. 9. The matrix demonstrates that the majority of samples are correctly classified, with values close to 1 along the diagonal. This indicates high classification accuracy for all classes.

Minor misclassifications are observed between visually similar disease classes, which is expected due to overlapping features. However, the overall performance remains strong, confirming the effectiveness of the proposed model.

F. Class-wise Performance Analysis

The model maintains consistent performance across all classes, including the minority class *Leaf_Rot*. The use of oversampling and WeightedRandomSampler ensures balanced learning and prevents bias toward the majority class.

G. Discussion

The experimental results demonstrate that the proposed hybrid model significantly improves classification performance

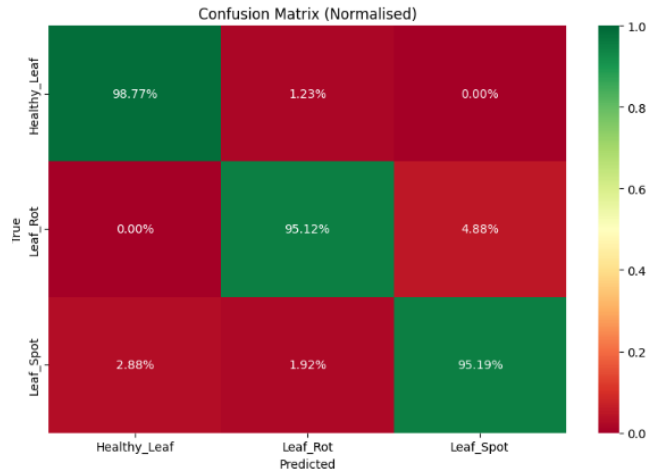

Fig. 9. Normalized confusion matrix showing class-wise prediction performance.

compared to traditional and single-model approaches. The integration of EfficientNet-B0 and Swin-Tiny enables the model to capture both local texture features and global contextual information.

Feature-level fusion plays a key role in enhancing representation learning by combining complementary strengths of CNN and transformer architectures. Additionally, data augmentation and class balancing techniques contribute to improved generalization.

The high ROC and PR scores indicate strong reliability across different thresholds, while the confusion matrix confirms accurate class-wise predictions. Furthermore, Grad-CAM analysis shows that the model focuses on disease-relevant regions, improving interpretability.

Overall, the results validate that the proposed hybrid fusion approach is effective, robust, and suitable for real-world betel leaf disease classification applications.

V. Conclusion

This study proposes a hybrid deep learning framework for betel leaf disease classification by integrating EfficientNet-B0 and Swin-Tiny through feature-level fusion. The proposed approach effectively combines the strengths of convolutional neural networks in capturing local texture features and transformer architectures in modeling global spatial relationships.

Experimental evaluation demonstrates that the model achieves high performance, with an accuracy of 97.07%, a Macro F1-score of 0.9597, and a Macro AUC of 0.9985. These results indicate that the fusion-based approach significantly improves classification accuracy and robustness compared to traditional and single-model methods.

The use of dataset balancing techniques ensures fair learning across all classes, while Grad-CAM enhances model interpretability by highlighting disease-relevant regions. These features make the proposed system both accurate and reliable for real-world agricultural applications.

In future work, the model can be extended by incorporating multimodal data such as environmental conditions or sensor inputs. Additionally, deployment in mobile or web-based platforms can enable real-time disease detection, making the system more accessible and practical for farmers.

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